iLearn Practice Exam

12BR · Chapter / Topic 14

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Topic 14: Communication and Internet Technologies

Name: _____

Class: _____

Date: _____

Time allowed: 40 minutes

TOTAL MARKS: 35

Marks obtained: ..... / 35

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 35.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 6 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, mark your work against the official Cambridge mark scheme (see the source list on the last page).
· Only 35 marks of past-paper questions are available for this chapter, so the recommended time is 40 minutes (1 min 1

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 **(a)** The Application Layer and Transport Layer are two layers of the TCP/IP protocol suite.

Describe the purpose of the Application Layer **and** the purpose of the Transport Layer.

Purpose of Application Layer .....

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Purpose of Transport Layer .....

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[5]

(b) Describe packet switching as a method of transmitting messages across the internet.

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..... [4]

2 Complete the following paragraph using the **most appropriate** words from the given list.

Some words are **not** used.

application	**controls**	**disconnects**	**four**
interconnects	**internet**	**link**	**lists**
many	**packets**	**seven**	**transport**

The TCP/IP protocol suite consists of ..... layers which can be thought of as a stack. It is designed to send ..... of data across the internet. It is also used as a communications protocol to ensure accurate delivery of data over private networks.

The lowest layer of the stack is the ..... layer. This is the layer that ..... nodes in the network.

[4]

3 **(a)** Identify **two** different layers of the TCP/IP protocol suite.

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..... [1]

(b) Describe how the TCP/IP protocol suite is applied when a message is sent through the internet from one host to another. Do **not** describe the function of individual layers of the TCP/IP protocol suite.

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..... [4]

4 Circuit switching may be used as a method of data transmission.

State **two** benefits and **two** drawbacks of circuit switching.

Benefit 1 .....

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Benefit 2 .....

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Drawback 1 .....

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Drawback 2 .....

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[4]

5 Draw **one** line to connect each TCP/IP layer to its **most appropriate** description.

TCP/IP layer	description
	ensures packets arrive in sequence without errors
Application	provides remote access to files on a host server to access and manage files stored in a remote location
Transport	interconnects nodes or hosts in a network, encapsulates IP packets into frames and ensures correct protocols are followed
Internet	identifies the intended network and host
Link	contains all the programs that exchange data and includes protocols: HTTP, FTP and POP3

[4]

6 **(a)** The Internet layer and Link layer are two layers of the TCP/IP protocol suite.

Describe the purpose of the Internet layer **and** the purpose of the Link layer.

Purpose of Internet layer .....

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Purpose of Link layer .....

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[5]

(b) Describe the function of a router in packet switching.

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..... [4]

END OF QUESTION PAPER

Source of each question (for marking)

Find the matching official Cambridge mark scheme for the series/paper shown, to mark your answers.

Q	Past paper	Original question	Marks
1	Oct/Nov 2025 Paper 31	Q3 (a),(b)	9
2	May/June 2026 Paper 32	Q2	4
3	Oct/Nov 2025 Paper 32	Q3 (a),(b)	5
4	Oct/Nov 2025 Paper 32	Q4	4
5	May/June 2026 Paper 31	Q4	4
6	Oct/Nov 2025 Paper 33	Q4 (a),(b)	9

Total: 35 marks